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Work Experience:

US Air Force, US Air Force Academy

US Air Force Academy
Air Force Academy, CO 80840 United States

06/2007 - 08/2009

Hours per week: 60

Assistant Professor, Instructor of Astronautics, Lieutenant Colonel USAF

Duties, Accomplishments and Related Skills:

Assistant Professor, course director, department personnel officer, and department accreditation lead. Created classroom demonstrations tools, test exams.

Supervisor: Colonel Martin France ((719) 333-4110)

US Air Force, National Reconnaissance Office, BPO/OCPA

14675 Lee Rd
Chantilly, VA 20151 United States

06/2014 – 06/2016

Hours per week: 65

Legislative Liaison & Senior Air Force Officer, Office of Congressional and Public Affairs, Business, Plans and Operations Directorate, Lieutenant Colonel USAF

Duties, Accomplishments and Related Skills:

Handpicked leader for the most challenging program within a joint DoD/CIA agency providing intelligence products to POTUS/Combatant Commanders/warfighters. Represents the DNRO to Congress and advises on technical, operational and policy issues for national space security. Leads daily interactions to defend & inform

budget submissions to senior Executive & Legislative branch leadership. Develops & executes corporate strategies to support Cabinet-level policy decisions to IC, DoD, OMB and Congress. Authors testimony, briefings & position papers in support of NRO's GEOINT Directorate covering a \$20B+ FYDP portfolio. Responsible for developing strategy for \$2B+ programs each year on Capitol Hill, which culminated in netted an additional \$203M+ for the NRO. O-8 appointed leader for 18 member unit and mentor championed unit to win 15 NAF personal and team awards. Piloted \$13B+ FY18 budget rollout to OMB/ODNI/USD(I) & Congress which secured the future of five ACAT-1 intelligence platforms. Organized and led 50+ Congressional engagements cementing Capitol Hill support for future GEOINT activities and reversed \$367M+ in cuts to GEOINT programs. Spearheaded a major reprogramming through Congress securing a \$76M+ booster and launch for a \$2.1B+ program three years ahead of need. Championed \$2B+ system plan through Congressional committees and reversed a \$46M+ fence to \$77M+ add to the program. Led appeal of NRO's top-priority systems with ODNI & JCS/J8 approval clearing a \$550M+ hurdle for a new resilient satellite program.

Supervisor: Ms Heidi Smith ((703) 808-1255)

US Air Force, National Reconnaissance Office, IMINT/ESPO

14675 Lee Rd

Chantilly, VA 20151 United States

06/2014 – 06/2016

Hours per week: 65

Chief, Intelligence, Surveillance and Reconnaissance Mission Payloads Branch, Major USAF

Duties, Accomplishments and Related Skills:

Handpicked leader for the most challenging program within a joint DoD/CIA agency providing intelligence products to POTUS/Combatant Commanders/warfighters. Leads a geographically and demographically diverse, 170+ industry and DoD lab team, integrating five developments into a \$330M advanced sensor on a \$10B+ ACAT-1D imaging satellite. Recognized as one of the top 5% Air Force officers in year group with a below the zone promotion to Lt Colonel. Whole person excellence maintaining exceptional fitness, award winning marathoner, private pilot, and active community leader. Built and maintains high functioning teams, at work and in the community utilizing skills in conflict resolution, trained team-building facilitator, interdepartmental athletics organizer, and go to spokesman for space, rockets, and astronomy motivational outreach. Directs 8-member payload branch utilizing leadership skills, dedication, and technical aptitude to balance cost, schedule, performance and risk through a multitude of design and manufacturing issues. Experienced software architect creating \$48M+ enterprise ground control system, leveraging traditional and cloud based tools including the MS Office suite and MS Project to build and manage timelines and resources, for a six-month early delivery. Full section manager with extensive experience in risk management, initiating contracts, system integration, program advocacy, design review, test and validation. Excels at hands-on

problem solving including: building a water rocket launcher for school and scout demonstrations, a homemade telescope, a ride-on hovercraft, rabbit hutches, a green house, home remodeling including porch and basement, and respectable home beer brewer.

- Team of the Year 2015, Team of the Quarter 2014 and 2015, Field Grade Officer of the Quarter 2015

Supervisor: Colonel Robert Introne ((703) 808-0212)

US Air Force, National Reconnaissance Office, IMINT/R&T

14675 Lee Rd

Chantilly, VA 20151 United States

08/2012 - 06/2014

Hours per week: 50

Program Manager, Advanced Imagery Systems, Major USAF

Duties, Accomplishments and Related Skills:

Proven program manager and system engineer with extensive experience in decision-making, strategic business planning, industry and government relations, financial and personnel management, cutting-edge research and development, and aerospace engineering. Leading 12 person joint DoD/CIA agency team integrating diverse systems into a \$267M+ electro-optical imaging payload for a \$10B+ program creating first-of-a-kind innovative satellite hardware solutions to satisfy warfighter intelligence requirements. Especially skilled at building effective, productive working relationships, and motivating teams to become the high performing Team of the Year directorate winner for 2013. Utilizing technical aptitude to learn, review, and solve technical issues on leading-edge hardware ensuring anomaly resolution and re-engineering on \$34M sensor failure eliminating program's top risk and averting four-month delay. Personally garnered NRO's top leadership approval for transition of technology demonstration program into major weapon system acquisition. Provided mentorship to young engineers and students alike through active community involvement and volunteer work. Leveraged skills in data management, modeling, and simulation including MS Excel, Visual Basic, Matlab, and specialized software tools, to organize development of a \$42M effort providing first new product delivery in 18 years and increasing imagery to warfighters by 166%

Supervisor: Colonel Timothy McKenzie ((703) 258-9272)

US Air Force, Air Force Institute of Technology/Civilian Institution

2950 Hobson Way

Dayton, OH 45433 United States

08/2009 - 08/2012

Hours per week: 60

PhD Candidate, University of Surrey, Surrey Space Centre, Captain USAF

Duties, Accomplishments and Related Skills:

Highly motivated, proactive, and applied student within a multi-national team completing a research based PhD program in Electronic Engineering at Surrey Space Centre. Invented and demonstrated analytic fuel-optimal control solutions for collision-free fuel-optimal reconfiguration for satellites in relative motion, demonstrated 2.1M times reduction in computation time and complexity over existing approaches allowing on-board computation reducing ground-based planning. Mastered programming skills in MatLab, C, C++, and Python to create and embedded model predictive control software for multi-satellite collaborative planning, maneuvering, and formation establishment. Skilled mentor, problem solver and time management instructor for fellow PhD candidates and the STRaND nano-satellite team. Worked hands-on building cube-sats and implementing code on microprocessors. Versatile international researcher, collaborator, technical speaker, and multinational research organizer across several countries including collaborative work with the MIT SPHERES and the DARPA F6 program solving challenging aspects of clustered formation flight, including implementing Clohessy-Wiltshire and J2 relative motion into flight algorithms for testing on board the International Space Station. Appointed systems analysis, mission planner, and astrodynamics design lead for collaborative EADs Astrium MEMs propulsion and Russian solar sail formation inspection mission.

Supervisor: Professor Phil Palmer (011 44 0770 975 9999)

US Air Force, US Air Force Academy

US Air Force Academy

Air Force Academy, CO 80840 United States

06/2007 - 08/2009

Hours per week: 50

Assistant Professor, Executive Officer, and Instructor of Astronautics, Captain USAF

Duties, Accomplishments and Related Skills:

Assistant Professor, course director, and faculty lead for \$230k student capstone design program, FalconLAUNCH. Air Force level winner and nominee for the 2008 Arthur S. Flemming Award for excellence in Applied Science, Engineering and Mathematics. Guided and worked hand-in-hand with students solving problems and building the first Academy rocket to reach space; setting undergraduate collegiate record of 106km in 2009 and flying a DoD SERB payload at hypersonic speeds. Enthusiastic educator responsible for instruction of 255 academy and foreign exchange cadets in Astronautical and Mechanical engineering.

Exclusively selected to instruct astronautics and space mission design course at French Air Force Academy, Teacher of the Quarter winner, 2008, and Company Grade Officer Teacher of the Quarter winner, 3 of 4 quarters of 2007. Academics and honor officer/mentor for 176 cadets providing academic career planning, ethical counseling, assisting with honor board investigations. Created partnership with White Sands Missile Range, Army, Navy, & Air Force test groups for on-site testing and launch of a two-stage sounding rocket. Orchestrated launch campaign including establishing communications, tracking, flight control, & range safety stations. Created classroom demonstrations tools, test exams, and update over 600 daily review questions.

- Awarded the Roger R. Bate Award for the Outstanding First Year Instructor of Astronautics 2007-2008.

- Awarded the John P. Wittry Award for top instructor of Engineering Design 2008-2009.

Handpicked executive officer for the Astronautics department responsible for the day-to-day operations of a 36-member military/civilian academic department. Personally responsible for ensuring effective communication between the dean (O-7), department head (O-6), faculty, and staff. Leveraged extensive attention to detail for flawless execution and successful completion of yearly performance reports, daily suspense tracking, unit inspections, and university re-accreditation. Executed hiring process for department personnel and created the first department wide continuity system.

Supervisor: Colonel Martin France ((719) 333-4110)

U.S. Air Force, Air Force Research Laboratory, Air Force Material Command

Kirtland AFB

Albuquerque, NM 87117 United States

01/2005 - 06/2007

Hours per week: 50

Program Manager, Space Superiority, Captain USAF

Duties, Accomplishments and Related Skills:

Architect and lead of a 25-member multi-divisional team developing a \$55 million space situational awareness technology demonstration program. Full spectrum manager of cost, schedule, and performance from concept development and requirements definition through critical design review. Accomplished program advocacy garnished the top priority mission within the directorate and a top five ranking of all Air Force Space Command experiments creating the agency's smallest satellite and first mission to geosynchronous orbit. Leveraged communication and management skills to write the concept of operations, acquisition documentation, proposals, industry days, and announcements. Experienced participant and organizer of source selections and contract negotiations. Selected to orchestrate 10-member/4-agency national team developing \$300 million space situational awareness demonstration to meet critical near-term needs and awarded Company Grade Officer of the Quarter, 3rd Quarter 2006. Spearheaded space data fusion center design, construction, and real-time satellite imagery streaming operations. Handpicked for elite team responsible for architecting organization's \$15M counterspace portfolio and long-range visions.

Supervisor: Dr. B. K. Singaraju ((505) 846-0583)

U.S. Air Force, Air Force Research Laboratory

Kirtland AFB

Albuquerque, NM 87117 United States

06/2004 - 01/2005

Hours per week: 50

Deputy, Advanced Spacecraft Mechanisms Group, 1Lt USAF

Duties, Accomplishments and Related Skills:

Deputy program manager for 8-member advanced spacecraft mechanisms group developing world-class research on mechanisms and launch vehicle vibration dampening. Responsible for 77-contracts, interfacing with 32 small businesses on innovative research with an annual budget of \$24M. Filled a GS-14 level program management position directly managing portfolios, personnel issues, and activity reports. Lead of a 10-person group responsible for design, fabrication, testing, delivery, and coordination of an innovative \$80M MDA ICBM shroud which cut average shroud mass by 75% and launch costs by \$6M. Developed and executed test plans, created schedules, and finalized configuration of \$2M miniaturized vibration isolation experiment integrated and flawless execution on the \$26M TacSat-2 satellite mission, launched 2006.

Supervisor: Dr Benjamin "Kyle " Henderson ((505) 853-6712)

US Air Force, Air Institute of Technology/Civilian Institution

2950 Hobson Way

Dayton, OH 45433 United States

06/2002 - 06/2004

Hours per week: 55

Draper Fellow, Charles Stark Draper Laboratory, 2Lt USAF

Duties, Accomplishments and Related Skills:

Professional student and researcher providing analysis and design of algorithms for guidance and control, attitude estimation, and extended Kalman filtering for NASA's SPIDR small satellite mission. Thesis work includes constellation design for mid-course ballistic missile defense applying astrodynamics and optimization

methods towards innovative satellite-based intercept constellation design and coverage definition. Received the “Vice President” award for best technical paper of 2004 within Draper Laboratory, Re-published in the Summer 2005 issue of the Draper Digest.

Education:

University of Surrey Guildford United Kingdom
Doctorate 08/2012

Major: Electrical Engineering

Relevant Coursework, Licenses and Certifications:

Research-based only program conducted at the Surrey Space Centre working with diverse team from 11 countries. Research focused on: Optimal control, Optimization methods, Astrodynamics, Orbital Proximity Operations for Small Spacecraft, and Embedded Programming. Dissertation work on Collision-free Optimal Reconfiguration of Small Satellites in Relative Motion. Specialized in analytic fuel-optimal planning and collision-free maneuvering for on-board computation. Developed technology for autonomous on-orbit collision identification, avoidance maneuvering and created approach for collision-free deployment of small satellite formations. Practical application of embedded algorithms demonstrated 2.1 million times reduction in computation time and complexity over existing approaches.

Massachusetts Institute of Technology Cambridge, MA United States
Master's Degree 06/2004

GPA: 4.8 of a maximum 5.0

Credits Earned: 72 Semester hours

Major: Aeronautics and Astronautics

Relevant Coursework, Licenses and Certifications:

Draper Fellow, Charles Stark Draper Laboratory

Relevant course work includes: Optimal control, Optimization methods, Stochastic estimation and control, Mathematics for engineers, and Astrodynamics. Thesis work on optimal satellite constellation design for mid-course ballistic missile intercept.

United States Air Force Academy Air Force Academy, CO United States
Bachelor's Degree 05/2002

GPA: 3.82 of a maximum 4.0

Credits Earned: 173 Semester hours

Major: Astronautical Engineering **Minor:** Mathematics **Honors:** Summa Cum Laude

Relevant Coursework, Licenses and Certifications:

Distinguished Graduate, 9th out of 943, on the Superintendent's List for Academic, Military, and Athletic Excellence.

Course work including: Astrodynamics, Dynamics and Controls, Satellite Engineering, Propulsion, Satellite Communications, Spacecraft and Mission Design, engineering mathematics (including mathematical modeling, multi-variable calculus, numerical integration, and differential equations), an emphasis on leadership training, as well as a diverse background of core classes and foreign language. Earned awards for top academic, military and physical performance including: Dean's List (8 of 8 semesters) receiving Outstanding Achievement awards for top performance in 13 classes, top graduate in major, and top performance in group level (150 peers) leadership.

- Chief Engineer for FalconSAT-2, Cadet built DoD satellite program for ionospheric modeling, designed to fly in the Shuttle as part of the NASA Hitchhiker program, led and motivated 34 cadets across multiple majors advancing a preliminary design to delivery of the fully tested flight unit in 10 months
- Summer Research with NASA Jet Propulsion Laboratory, 2001, Mission Engineering and Planning Group, helped mission designers in Mars mission group maximize communication on the Deep Space Network
- "Moore Award" winner, USAF Academy Engineering Division, for Research for research conducted at NASA JPL
- "Outstanding Cadet in Astronautical Engineering for the Graduating class of 2002", number one graduate in major with a 4.0 major's GPA and top performance in all major's courses.
- Top Gun award winner for leadership position in military squadron
- Received the highest military recognition for work in each Element leader role and as the BCT first Sergeant
- "Space Grant Session Award" winner at the Space Grant Conference in Boulder, Colorado, April 2002.

Job Related Training:

- Acquisition Leadership Challenge Program-I, February 2014
- Air Command and Staff College, In-Residence/Correspondence Completion, 2012
- Space 200, Space Professional Level-2 certification, NSSI 2008
- Squadron Officer's School In-Residence/Correspondence, Top-Third Graduate (Used in place of D.G. at the time), Top-Flight Overall for best all-around performance
- Acquisitions Fundamentals, Wright-Patterson AFB, 2005
- Acquisition Professional Development certifications: Engineering Level-3, Program Management Level-2, Test Level-1
- CPR certified
- Completed SCUBA instruction at the US Air Force Academy, 2002
- Completed Basic Freefall parachuting instruction at the US Air Force Academy, 2002
- Licensed Private Pilot, Completed 1997, member of US Air Force Academy Flying team

Language Skills:

Language	Spoken	Written	Read
Spanish	Novice	Novice	Novice

Affiliations:

American Institute of Aeronautics and Astronautics - Member
 Boy Scouts of America - Den Leader
 Sigma Gamma Tau - Member
 Tau Beta Pi - Member

Professional Publications:

- Primary author of "Analytic Model Predictive Controller for Collision-Free Relative Motion Reconfiguration," AIAA Journal of Guidance, Control, and Dynamics, July-August 2012
- Primary Author of " Onboard Semianalytic Approach to Collision-Free Formation Reconfiguration," IEEE Transactions on Aerospace and Electrical Systems, July 2012
- Primary Author of "Path Planning for Fuel-Optimal planning and Collision-Free Formation Flying Trajectories," Microelectronics Reliability, 2011
- Contributing Author of "Formation Deployment & Separation Simulation of Multisatellite Scenarios using SatLauncher," Microelectronics Reliability, 2011
- Primary author of "Path Planning for Fuel-Optimal Collision-Free Formation Flying Trajectories," IEEE Aerospace Conference, Big Sky, MT, 2011.
- Primary author of "Finite Element Approach with Analytic Enhancement for Onboard Fuel-Optimal Collision-Free Formation Reconfiguration," 8th International ESA Conference on Guidance, Navigation, and Control Systems, Karlovy Vary, Czech Republic, June 2011
- Contributing author of, "Small Satellites: Past, Present, and Future," AIAA, Jan 2009
- Contributing author of, Aeroelastic Analysis and Optimization of FalconLAUNCH Sounding Rocket Fins," 47th AIAA Aerospace Sciences Meeting, 2009
- Presented classified paper at the AFRL Maui Optical Site Conference, 2006.
- Primary author of "An Improved Kalman Filter for Satellite Orbit Predictions"; Published in the Journal of Astronautical Sciences July-Sep 2005 Issue.
- Presented classified paper at the MIT/LL Space Control Conference, 2005.
- Primary author of AAS 04-295, "Constellation Design for Mid-Course Ballistic Missile Intercept," AAS/AIAA Space Flight Mechanics Conference February 2004, Maui, HI. Research completed at Draper Laboratory/M.I.T.
- Author of student paper, "Improving on the Extended Kalman Filter"; Presented at the Space Grant Conference, Boulder, CO, April 2002.

- Primary author of AIAA-2001-4776, “Mars Surface Direct Communications”; Research completed at JPL; Presented at the AIAA Space Conference and Exposition 2001, Albuquerque, NM.
- Author of AIAA-2001-4774, “Learning Space by Doing Space—The USAF Academy's FalconSAT Program”; Presented at the AIAA Space Conference and Exposition 2001, Albuquerque, NM.
- Author of IEEE-510, “Bounding the Problem: Microsatellite Design Using Commercial-Off-The-Shelf Architecture”; Presented at the IEEE Aerospace Conference, Big Sky, MT, March 2001.

References:

Name	Employer	Title	Phone	Email
Aaron Weiner(*)	US Air Force, SAF/AQ	Lt Col	(719) 687-6362	aaron.m.weiner.mil@mail.mil
Todd Nathaniel	US Air Force, Air Force Academy	Lt Col	(703) 867-4515	nathanta@me.net
Dr. Phil Palmer(*)	University Of Surrey, Surrey Space Centre, United Kingdom	Professor	011-44-0770 975 9999	philssca@gmail.com
Martin France (*)	US Air Force, Air Force Academy	Colonel, Department Head	(719) 333-4110	Martin.France@usafa.edu
Peter Norton	US Air Force	Lt Col	(571) 315-1278	peter.norton648@gmail.com

(*) Indicates professional reference

Additional Information:

COMMUNITY OUTREACH AND INVOLVEMENT

- Cub Scout Den Leader, Assistant Pack leader, and Pack Leadership Council Member, leading meetings, activities, outings, summer camps and responsible for 64 kids across four grades. Organizer of annual model rocket building and pack launch competitions
- Classroom volunteer providing introduction to space, rockets, and astronomy lessons. Organized school wide student water-rocket launches with egg payload recovery
- Head Coach for 12U Girls Softball coach
- Home Owners Association (HOA) Board member and neighborhood advocate, responsible for board meetings, neighbor conflict resolution, HOA newsletter, minutes, complaints, and architectural change requests. Organized neighborhood Easter egg hunt, community picnic, and annual Christmas light contest. Managed neighborhood \$7k restoration projects, organized competitive bids, contract negotiations, and monitored completion.
- Active community volunteer for Girl Scouts complete their Bronze Award Service Project, 4H community projects, Kenneth Culbert Elementary PTA member & volunteer

PROGRAMMING SKILLS & SECURITY CLEARANCE:

- Significant experience in several programming languages including: Ada, C, C++, HTML, Java, J++, & Matlab/Simulink, Python, Visual Basic,
- TS-SCI clearance, SSBI 12 Nov 2013